

- Activated B-lymphocytes grow in size and then divide by mitosis. Many further mitotic cell divisions occur, increasing the number of B-lymphocytes with receptors specific to the antigen. Eventually, cells produced in this process will develop into either plasma cells that secrete antibodies or memory B-cells.

The diagram illustrates the process of B-lymphocyte activation and differentiation. It begins with a B-lymphocyte that has not yet been activated, which binds to an antigen via its receptor. This is followed by cell growth and activation, leading to mitosis. The resulting cells undergo repeated cycles of cell division by mitosis, forming a large population of activated B-lymphocytes. These cells then differentiate into two types: plasma cells that produce and secrete antibodies, and memory B-cells.

antigen

receptor

B-lymphocyte that has not yet been activated

binding of antigen and activation of B-lymphocyte

cell growth

mitosis

repeated cycles of cell division by mitosis

develops into

antibodies

plasma cell that produces and secretes antibodies

memory B-cell

The development of plasma cells and memory B-cells in this process depends on transcription factors.

- [3]

- (b)** Transcription factors are proteins. Genes that code for proteins can become mutated.

Describe how different types of gene mutation can cause changes in the protein that is synthesised.

..... [5]

- (c)** BLIMP-1 is a transcription factor that is essential for the development of plasma cells and memory B-cells in the process shown in Fig. 3.1. BLIMP-1 reduces the synthesis of c-Myc in B-lymphocytes. c-Myc is a protein that is required for the mitotic cell cycle to continue.

Suggest **and** explain how a mutation in the gene coding for BLIMP-1 can prevent the development of plasma cells and memory B-cells.

.....[3]

(d) Microarrays can be used to analyse the effect of transcription factors, such as BLIMP-1, on gene expression.

(i) Describe how a microarray is used in the study of gene expression.

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..... [4]

(ii) BCL6 is another transcription factor found in B-lymphocytes. The effect of BCL6 on gene expression was compared in two samples of B-lymphocytes.

- Sample 1 consisted of B-lymphocytes that were producing BCL6.
- Sample 2 consisted of B-lymphocytes that were **not** producing BCL6.

Suggest why a microarray is suitable for identifying the function of the transcription factor BCL6 in these two samples.

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..... [2]

[Total: 17]

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